PROVISIONAL PATENT APPLICATION

AUTHENTICATING A WRITER BY DERIVING KEY MATERIAL FROM A DETERMINISTIC INK SIMULATION OF HANDWRITTEN INPUT ON A PUBLIC REFERENCE SURFACE

Inventor: Tabor Holly, Oregon, United States

Filed under 35 U.S.C. § 111(b). Related to the applicant's co-pending provisional application "Handwriting Authentication Using a Writing Surface Derived from a Cryptographic Key and a Deterministic Ink Simulation."

TECHNICAL FIELD

[0001] This disclosure relates to user authentication, behavioral biometrics, and computer graphics simulation, and in particular to recognizing a writer from the dynamics of their handwriting by passing the handwriting through a deterministic ink-and-paper simulation on a fixed public reference surface, and using the result, together with a password, to unlock a cryptographic key.

BACKGROUND

[0002] Handwritten signature biometrics extract features such as stroke velocity, pressure, and timing from captured pen samples and compare them to an enrolled template. Such systems face two persistent difficulties. First, feature design is ad hoc: the features chosen determine what is measured, and a forger who can trace the visible shape of a signature may defeat shape-dominated features. Second, templates go stale: a person's handwriting varies with mood, fatigue, and age, so a template captured at enrollment progressively fails to recognize its owner, and systems must choose between lockouts and loose tolerances.

[0003] Fuzzy extractors are known constructions that derive a stable cryptographic value from a noisy input using public helper data, provided successive inputs are within a bounded distance. Their usefulness for handwriting has been limited by the low entropy and instability of hand-designed features.

[0004] The applicant's co-pending application describes a deterministic ink simulation in which stroke samples deposit ink onto a procedurally generated writing surface and the ink flows, settles, and dries under fixed integer rules, such that the settled ink depends on both the stroke dynamics and the surface and is byte-for-byte reproducible. The present disclosure uses that simulation for a different purpose: as the measuring instrument for the writer's hand.

SUMMARY

[0005] Handwritten input is rendered through the deterministic ink simulation on a public reference surface that is identical for all users. Because the simulation is a fixed physical transform, the settled ink amplifies exactly the dynamic traits that a visual forgery lacks: dwell produces beads and dark rings; speed produces starved hairlines; pressure produces width; rhythm produces drying boundaries; slant produces a characteristic landing of hairlines on the surface fibers. The settled facts are quantized into a bit vector, the hand measure. The simulation thus serves as the feature transform, replacing hand-designed features with a physical one.

[0006] A fuzzy extractor with per-account helper data converts the hand measure into stable hand bits. The hand bits are combined with a password through a slow key derivation function to decrypt a stored blob containing a randomly generated account private key. The private key is never derived from the hand, so changes in handwriting never change the key; the hand and password merely open it. Neither the hand alone nor the password alone can open it.

[0007] Authentication is performed by writing a secret phrase chosen by the writer and typing the password. The phrase is secret because the writer's other handwriting, including its dynamics, may be public, and a hand measure computed from public writing would otherwise be computable by anyone; the secret phrase ensures the measure depends on content only the writer knows. The stroke samples of the phrase never leave the device; only a sealed digest and a server-issued nonce are sent, which prevents replay. The device recovers the key, signs the nonce, and the server verifies the signature against the account's public key. Each successful authentication is used to re-sketch the matching helper data, so that enrollment is continuous and follows drift. Multiple helper sketches per account accommodate distinct writing rhythms (including different input devices), and a wide-tolerance sketch built from a full page of writing, also kept private, serves as a backstop when the phrase fails.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates derivation of the hand measure from stroke samples via the simulation on the reference surface.

[0009] FIG. 2 illustrates recovery of the account private key from the hand measure, helper data, and password.

[0010] FIG. 3 is a flow chart of authentication.

[0011] FIG. 4 illustrates continuous re-enrollment and multiple rhythm sketches.

DETAILED DESCRIPTION

Hand measure (FIG. 1)

[0012] Referring to FIG. 1, stroke samples 402 comprising position, timestamp, and optionally pressure are supplied to the deterministic ink simulation 404 of the co-pending application, configured with a public reference surface 406 generated from a fixed, published seed so that every device and every account uses the identical surface. The simulation produces settled facts 408: for each cell, the settled pigment, and optionally derived features such as bead rings and drying boundaries. A quantizer 410 maps the settled facts to a bit vector 412, the hand measure. In one embodiment the quantizer partitions the rendered region into zones aligned to the phrase's stroke bounding boxes and, for each zone, thresholds aggregate quantities such as total settled pigment, ring count, maximum bead height prior to settling, hairline minimum width, and the orientation histogram of fiber-carried ink. The quantizer is part of the versioned simulation law and is deterministic.

[0013] Because the simulation responds to dwell, speed, pressure, and rhythm, two writings with identical visible shape but different dynamics (an original and a tracing) yield different settled facts and different hand measures, while repeated writings by the same hand yield measures within a bounded distance.

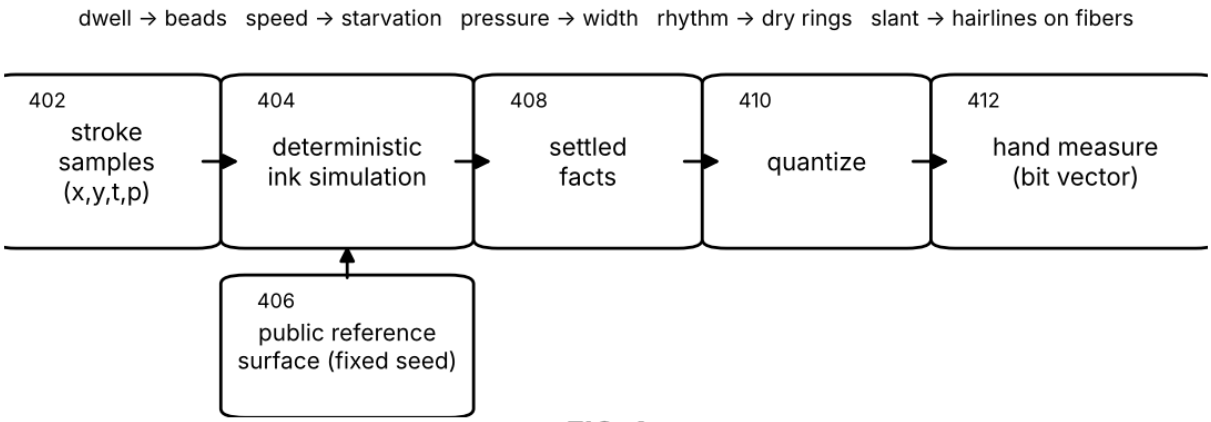


FIG. 1

Key recovery (FIG. 2)

[0014] Referring to FIG. 2, at registration a random account private key 432 is generated on the device. The user writes a page; its hand measure is used to construct helper data 420 of a fuzzy extractor 422 (for example a secure sketch based on an error-correcting code) together with random hand bits 424. The hand bits and the user's password 426 are passed through a slow key derivation function 428, and the result encrypts the private key into a blob 430 stored on a server with the helper data and the public key. The private key, the hand bits, and the password are never transmitted.

[0015] At authentication a fresh hand measure 412 and the helper data 420 are supplied to the fuzzy extractor, which reproduces the hand bits 424 exactly if the measure is within the code's correction

radius, and otherwise produces unrelated bits. The reproduced hand bits and the typed password are passed through the key derivation function, and the blob is decrypted to recover the private key. A wrong hand or a wrong password yields a wrong decryption key, and the blob fails to decrypt.

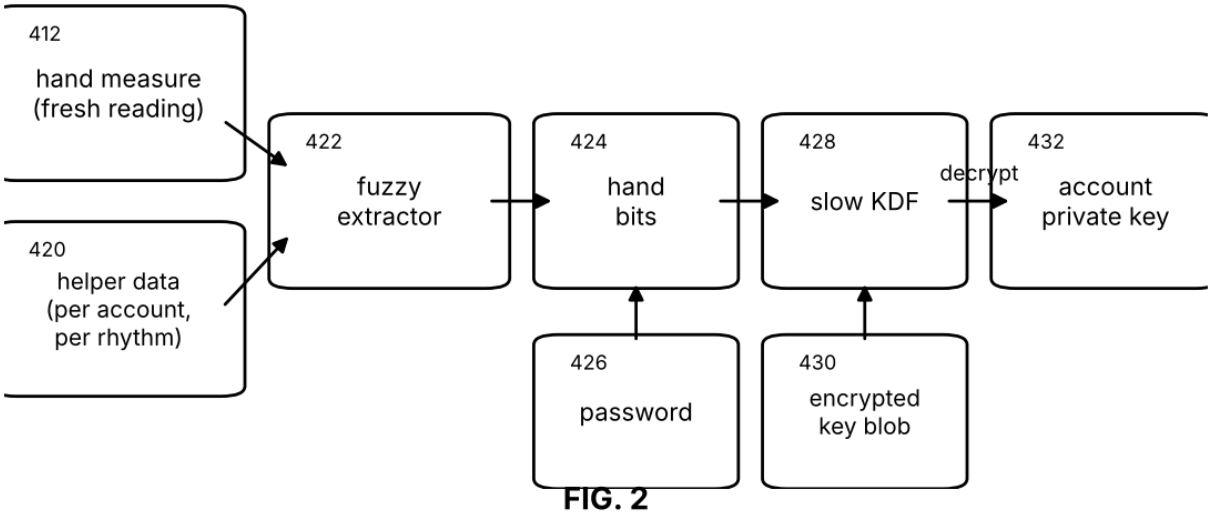


FIG. 2

Authentication (FIG. 3)

[0016] Referring to FIG. 3, a server issues 440 a nonce. The writer writes their secret phrase and types the password 442. The device derives the hand measure, recovers the hand bits and the private key 444, signs the nonce, and seals a login record containing the nonce and a digest of the stroke samples 446; the stroke samples themselves are not transmitted. The server verifies the signature and the record against the account public key 448 and the single-use nonce. The device then re-sketches the matching helper from this reading 450. Guessing is made online-only by evaluating the key derivation through a server-held oblivious pseudorandom function (for example RFC 9497), so every candidate requires a server round trip that is counted and rate-limited per account and per network address; a captured login record cannot be re-submitted because its nonce is spent.

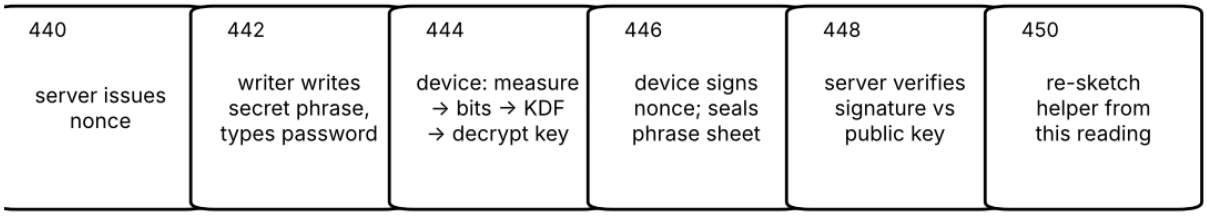


FIG. 3

Continuous enrollment and rhythms (FIG. 4)

[0017] Referring to FIG. 4, each successful authentication 460 yields a fresh measure 462. If the measure matches an existing sketch i, that sketch is refreshed from the new measure 464. If it matches no sketch but the password was correct, a new sketch is added 466, up to a configured number N of sketches 468, each corresponding to a distinct writing rhythm. Enrollment is therefore continuous: the hand that opens the account is the hand as it is now, not as it was at registration. In one embodiment two classes of sketch are kept: a phrase sketch with tight tolerance for routine authentication and a page sketch with wide tolerance, built from a full page of writing, used as a backstop when the phrase sketch fails.

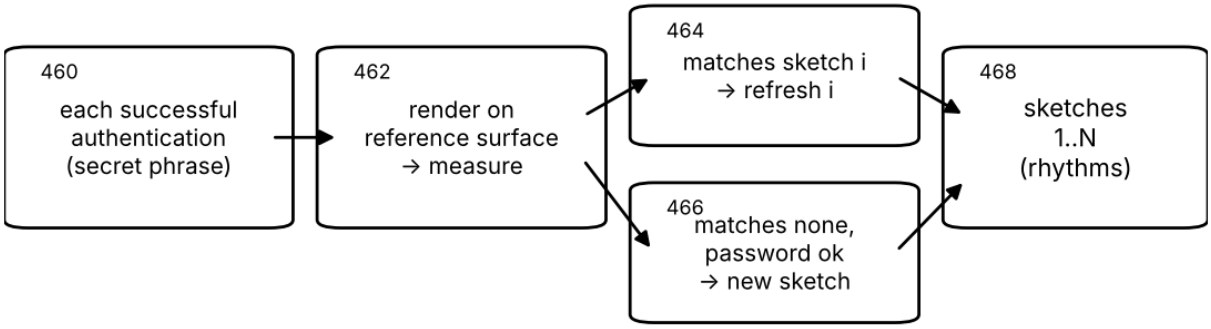


FIG. 4

Security properties

[0018] The hand measure contributes an entropy on the order of tens of bits; the password contributes further entropy; the slow key derivation function and server-side rate limiting make online guessing impractical and offline guessing costly. The password alone cannot open the blob because the hand bits are required; the hand alone cannot because the password is required. The private key is random and independent of the hand, so the account remains the same cryptographic identity across all changes of handwriting. Because the simulation is deterministic and the reference surface is public, the hand measure can be computed on any device and no biometric template in the conventional sense is stored; only helper data from which the measure cannot be recovered.

Relation to the co-pending application

[0019] The account private key recovered by this method is the private key of the co-pending application: it derives per-sheet seeds, grows the writer's own surface, and seals records. The two methods compose: the hand opens the key, and the key grows the paper on which the hand's writing is rendered and sealed.

CLAIMS

What is claimed is:

1. A method of authenticating a writer, comprising: receiving a sequence of stroke samples of handwritten input; applying a deterministic simulation that deposits ink in response to the stroke samples onto a reference writing surface generated from a published seed and propagates the ink among cells of the surface in dependence on cell states, to produce settled ink facts; quantizing the settled ink facts to produce a hand measure; applying a fuzzy extractor to the hand measure using stored helper data to produce hand bits; deriving a decryption key from the hand bits and a password; decrypting a stored blob with the decryption key to recover a private key; and producing, with the private key, a digital signature over a challenge issued for the authentication.

2. The method of claim 1, wherein the private key is generated randomly at registration independently of the handwritten input, such that changes in the writer's handwriting do not change the private key.

3. The method of claim 1, wherein the handwritten input is a phrase chosen by and known only to the writer, the stroke samples of the phrase are not transmitted from the writer's device, and the digital signature covers a single-use nonce issued by a server, such that a hand measure cannot be computed from the writer's other published handwriting and a recorded authentication cannot be replayed.

4. The method of claim 1, wherein the simulation is performed in fixed-point integer arithmetic with a fixed rule order and the reference surface is identical for all writers, such that the hand measure of given stroke samples is identical on any conforming device.

5. The method of claim 1, wherein the quantizing comprises deriving, for regions of the settled ink, one or more of: total settled pigment, a count of drying rings, a maximum ink height prior to settling, a minimum hairline width, and an orientation distribution of fiber-carried ink.

6. The method of claim 1, further comprising, after each successful authentication, re-computing the helper data from the hand measure of that authentication, whereby enrollment follows changes in the writer's handwriting.

7. The method of claim 1, wherein a plurality of helper data sets is stored for the writer, the fuzzy extractor is applied with each until one produces hand bits that decrypt the blob, and a hand measure that matches no stored helper data set while the password is correct causes a new helper data set to be stored.

8. The method of claim 1, wherein a first helper data set is constructed from a short phrase with a first correction radius and a second helper data set is constructed from a page of writing with a larger correction radius, and the second is used when the first fails.

9. The method of claim 1, wherein the private key is further used to derive a per-sheet seed by signing a sheet identifier, to generate a writer-specific writing surface from the seed, and to seal records of the writer's handwriting rendered on the writer-specific surface.

10. The method of claim 1, wherein deriving the decryption key comprises evaluating an oblivious pseudorandom function with a server-held per-account secret over a blinded combination of the hand bits and the password, such that each candidate decryption key requires a server evaluation that the server counts and rate-limits, and the hand bits, the password, and the private key are not transmitted.

11. A system comprising a writer device and a server configured to perform the method of claim 1.

12. A non-transitory computer-readable medium storing instructions that, when executed, cause one or more processors to perform the method of claim 1.

ABSTRACT

A writer is authenticated by passing handwritten stroke samples through a deterministic ink simulation on a public reference writing surface, so that the settled ink serves as a physical feature transform amplifying dwell, speed, pressure, rhythm, and slant. The settled facts are quantized into a hand measure; a fuzzy extractor with per-account helper data converts it to stable hand bits; the hand bits and a password derive a key that decrypts a stored blob holding a randomly generated account private key, which signs a fresh challenge. The key is independent of the hand, so handwriting changes never change the identity. The phrase is secret and its strokes never leave the device, so the measure cannot be computed from the writer's public handwriting; a server-held oblivious pseudorandom function makes guessing online-only. Helper data is re-sketched from each authentication, and multiple sketches accommodate distinct writing rhythms and devices, so enrollment is continuous. Neither hand nor password alone suffices.